

Robot Navigation and Tracking System Documentation

Introduction

This document provides an overview of the Robot Navigation and Tracking System Interface, which is designed to assist in tracking and navigating an agriculture robot.

The system consists of three main parts:

1. **Home Page – Choose Pathway or Live Tracking**
2. **Set Pathway – Set Area for the Robot**
3. **View Live Tracking – Monitor Robot's Movement**

Each part plays a crucial role in either setting up the robot's navigation or tracking its movements. The system utilizes Google Maps for location visualization and Firebase for real-time GPS updates.

Part 1: Home Page – Choose Pathway or Live Tracking

Description

The home page serves as the initial interface for the user to select between two primary actions:

- **Set Pathway:** This option leads the user to a page where they can define and configure the robot's route.
- **Live Tracking:** This option allows the user to view the robot's live position and track its movements in real-time.

The user interface is styled using Bootstrap to ensure a clean and responsive layout with two buttons to select between setting a pathway or tracking the robot.

Key Components:

- **HTML Layout:** Utilizes a simple layout with buttons to navigate between the two actions.
- **Navigation Links:** Directs the user to the relevant pages for pathway setting or live tracking.
- Background image with Industry 4.0 logo

Part 2: Set Pathway – Choose Area for the Robot

Description

This part of the system allows the user to define a specific area or pathway for the robot to travel. The map is displayed using Google Maps, and the robot's GPS coordinates are retrieved from Firebase to update its position in real-time.

The user can visualize the robot's movement along its defined path as GPS coordinates are continuously updated. This provides an interactive way to monitor and adjust the robot's route.

Key Components:

- **Google Maps API:** Used to display the map and the robot's position.
 - **Firebase:** Fetches real-time GPS data for the robot, allowing its position to be updated live on the map.
 - **Marker and Polyline:** The robot's position is represented by a marker on the map, and its path is traced using a polyline.
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Part 3: View Live Tracking – Monitor Robot's Movement

Description

In this step, the user can monitor the robot's real-time position and the path it follows. This page continually updates the robot's location using GPS data fetched from Firebase and visualizes it on Google Maps. The map refreshes every time the robot moves, ensuring accurate live tracking.

Key Components:

- **Real-Time GPS Updates:** The system fetches the latest GPS data from Firebase, ensuring the robot's location is always up-to-date.
 - **Google Maps API:** Displays the robot's location on the map and updates it as new data is received.
 - **Path Visualization:** The robot's movement is traced by a polyline, allowing the user to see its exact route.
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Conclusion

Key Technologies Used:

- **Firebase:** A cloud-based real-time database used to store and retrieve the robot's GPS coordinates, ensuring continuous updates.
- **Google Maps API:** Provides the interface for displaying the robot's position and its path on a map.
- **Bootstrap:** Used for responsive and visually appealing user interface design.

Next Steps:

- Users can further customize the system to add features such as different routes, obstacle detection, and multiple robot tracking.
- The system is scalable and can be enhanced with additional data points related to the robot's environment or performance.

GitHub Repository:

All source code, including HTML, JavaScript, and CSS files, is available in the project's GitHub repository.